



ASSIGNMENT - 2

BUSINESS INTELLIGENCE - BM234

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1. Introduction

A bank ran a personal loan campaign across 5,000 customers. Only 480 of them said yes, which is an acceptance rate of 9.6%. Reaching out to every customer costs money, and most of those calls and messages do not turn into loans. So the bank wants to know who is most likely to accept, and focus the next campaign on them.

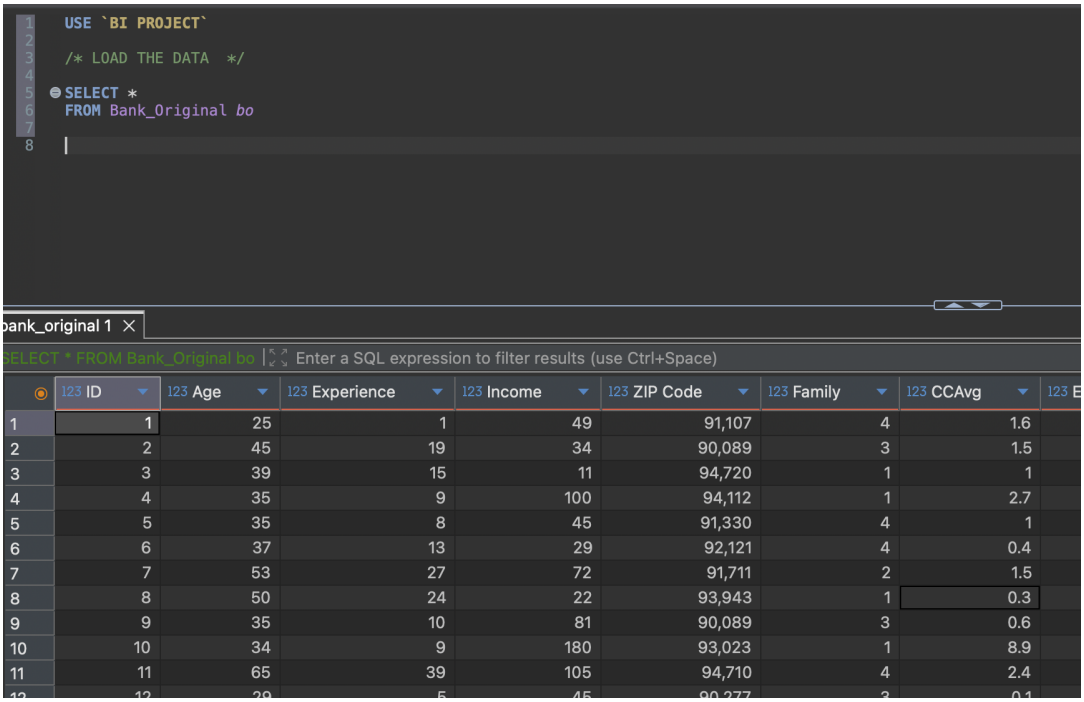
We used SQL to check, clean and analyse the data, and Power BI to turn the results into an interactive dashboard. We set out to answer five questions:

1. What share of customers accepted the loan overall?
2. How do income and education affect acceptance?
3. Do banking habits (CD account, online banking, credit card, card spending) matter?
4. Do personal factors such as age and family size matter?
5. Which customer segment should the bank target next?

2. Dataset Description

The dataset has 5,000 customer records from the earlier campaign. Each row is one customer. The target column is **Personal Loan**: 1 means the customer accepted the loan, 0 means they did not. Income, card spending and mortgage values are in thousands of dollars.

Column	Meaning
customer_id	Unique customer number
age	Customer age in years
experience	Years of professional experience
income	Annual income ($K)
zip_code	Home ZIP code
family	Family size (1 to 4)
ccavg	Average monthly credit card spending ($K)
education	1 = Undergraduate, 2 = Graduate, 3 = Advanced/Professional
mortgage	Value of house mortgage ($K)
personal_loan	Accepted the loan in the last campaign (target)
securities_account	Has a securities account with the bank
cd_account	Has a certificate of deposit account
online_banking	Uses internet banking
credit_card	Uses a credit card issued by the bank



	123 ID	123 Age	123 Experience	123 Income	123 ZIP Code	123 Family	123 CCAvg
1	1	25	1	49	91,107	4	1.6
2	2	45	19	34	90,089	3	1.5
3	3	39	15	11	94,720	1	1
4	4	35	9	100	94,112	1	2.7
5	5	35	8	45	91,330	4	1
6	6	37	13	29	92,121	4	0.4
7	7	53	27	72	91,711	2	1.5
8	8	50	24	22	93,943	1	0.3
9	9	35	10	81	90,089	3	0.6
10	10	34	9	180	93,023	1	8.9
11	11	65	39	105	94,710	4	2.4

Figure 1: The raw table in DBeaver (first rows of Bank_Original).

3. Tools and Method

Tool	Purpose
MySQL and DBeaver	Loading the data, cleaning it in a SQL view and running the analysis queries
Power BI Desktop	Building the three-page interactive dashboard
GitHub	Version control and project documentation

Our workflow was:

1. Load the raw CSV into a MySQL table.
2. Build a clean view, `vw_customers`, that fixes the data problems and turns codes into readable labels.
3. Run the analysis queries on that view.
4. Export the clean view and the top segments result to CSV.
5. Load the CSV files into Power BI and build the dashboard.

4. Data Cleaning

We checked the raw table before doing any analysis. Three quick queries found 52 rows with negative experience, no duplicate customer IDs, and one ZIP code that was not five digits long.

Check	Result	What we did
Rows with negative Experience	52	Fixed with ABS in the view
Duplicate customer IDs	0	Nothing to fix
ZIP codes not 5 digits long	1	Padded with a leading zero
Unused columns (Column15 to Column20)	6	Left out of the view

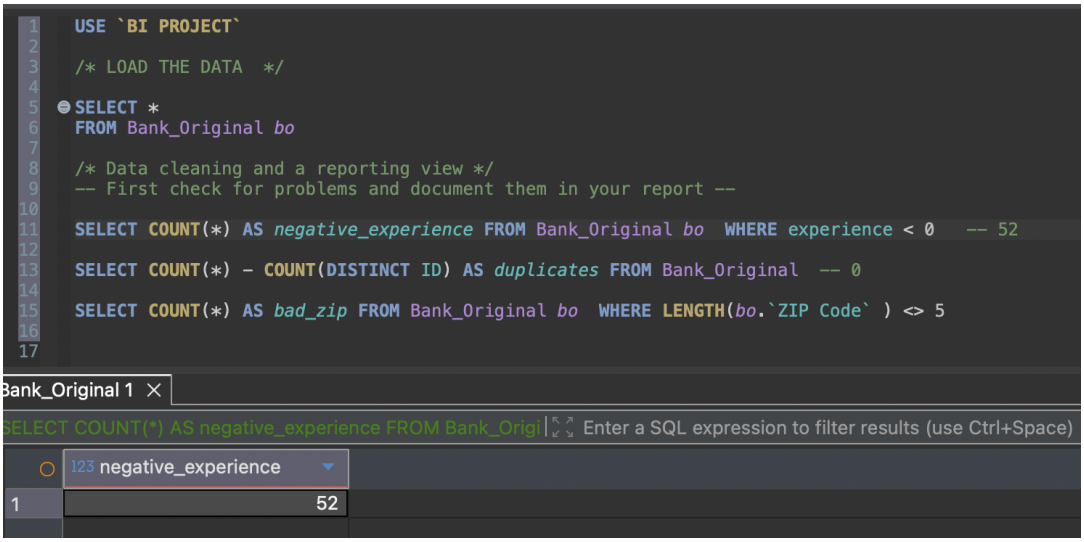


Figure 2: 52 rows have negative Experience.

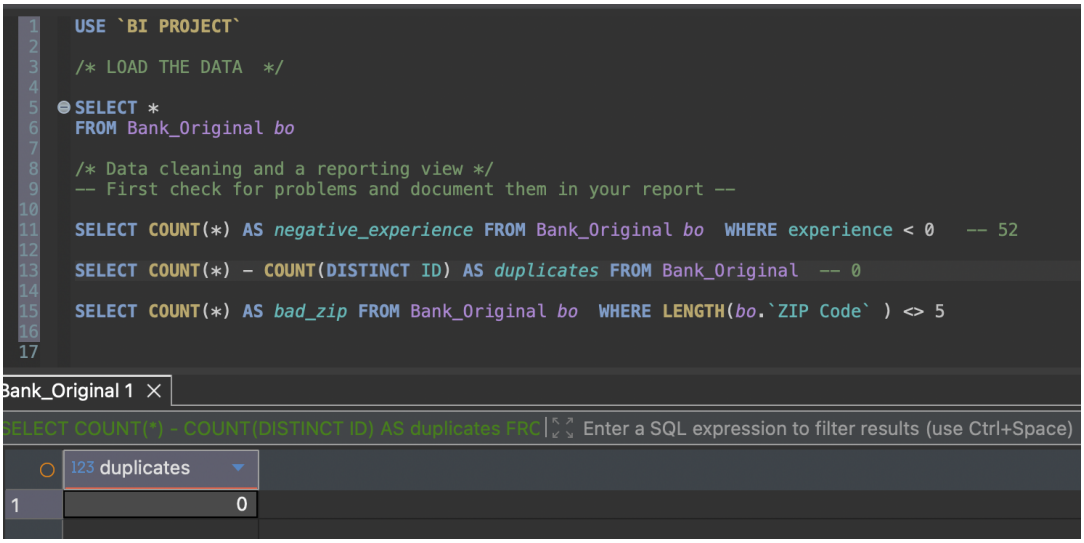


Figure 3: No duplicate customer IDs.

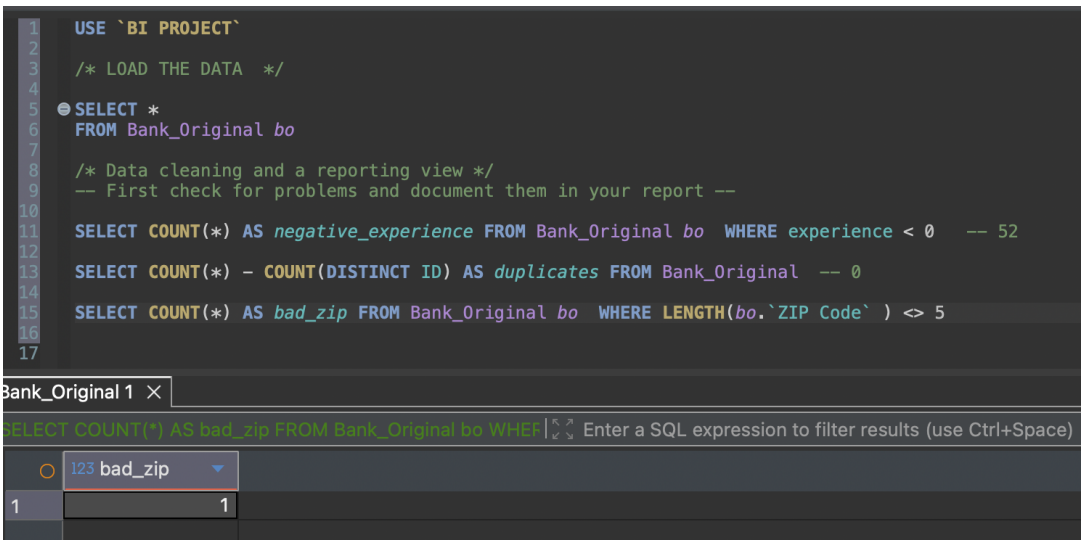


Figure 4: One ZIP code is not 5 digits long.

Everything was fixed inside a SQL view, so the original table stays untouched. The view does the following:

- **Negative experience:** corrected with the ABS function.

- **ZIP codes:** padded to five digits with LPAD. In Power BI we also set zip_code to Text so the leading zero is not lost again.
- **Coded values:** education 1/2/3 became Undergraduate/Graduate/Advanced-Professional, and personal_loan 1/0 became Accepted/Not Accepted. The yes/no columns (securities account, CD account, online banking, credit card) became Yes/No.
- **Bands:** we added age_group, income_band and spend_band columns so results can be grouped clearly.

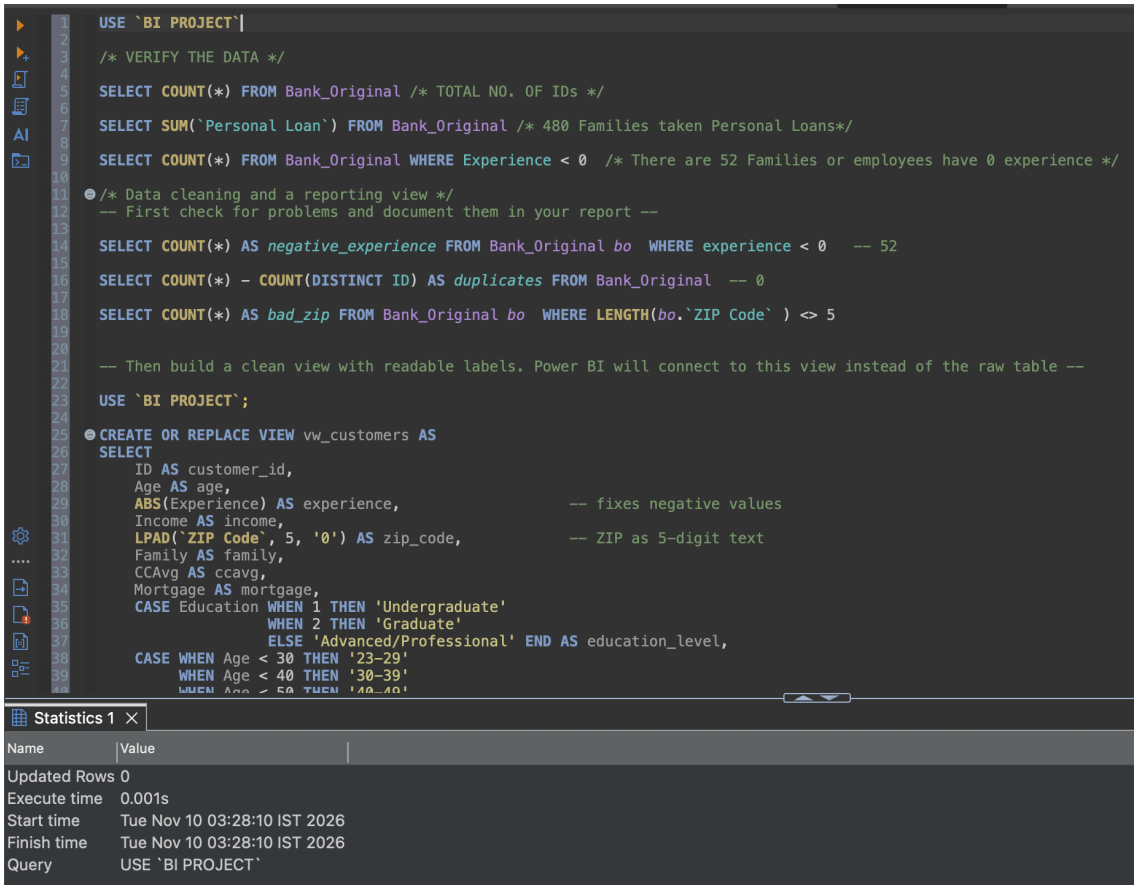


Figure 5: The data checks and the start of the CREATE VIEW statement.

After building the view we checked it against the raw data. It still has 5,000 rows and 480 accepted loans, and the smallest experience value is now 0. Nothing was lost and no negative values are left.

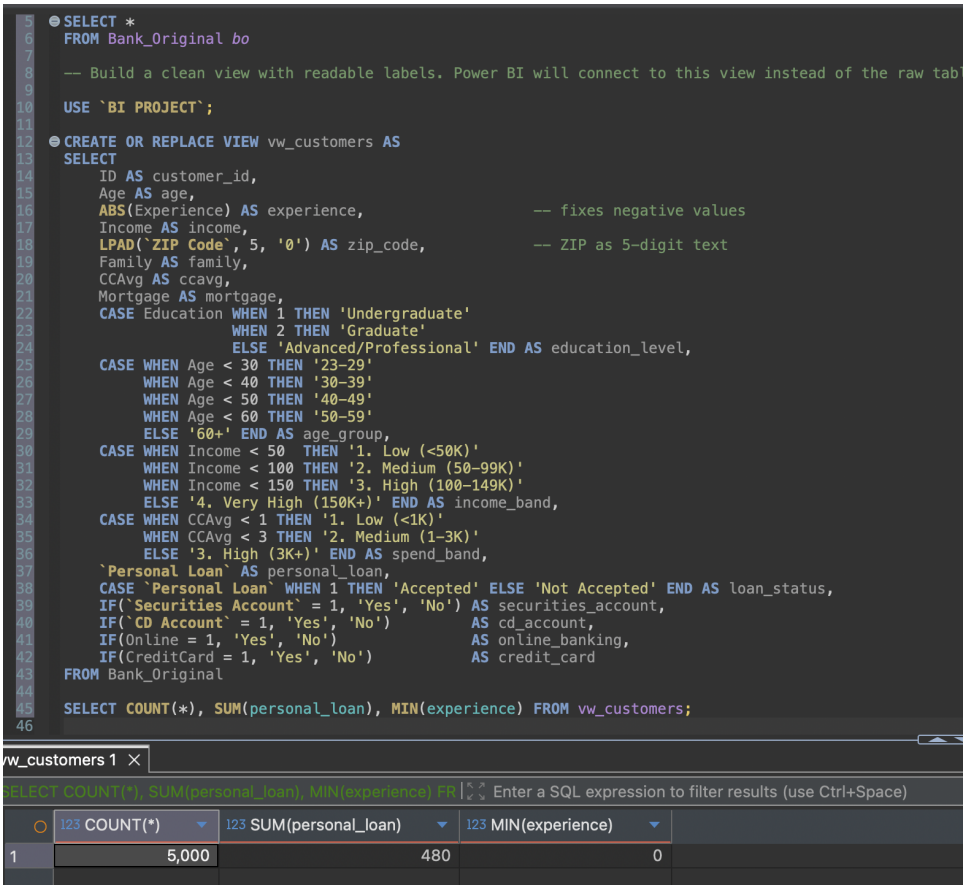


Figure 6: The full view definition and the check: 5,000 rows, 480 loans, minimum experience 0.

5. SQL Analysis

We ran ten queries on the clean view. The first six look at one factor at a time. The last four find the best customer segment and test whether it holds up. The online banking and credit card results come from the dashboard rather than from these queries.

Query 1: Overall results

Total customers	Loans accepted	Acceptance rate	Average income ($K)
5,000	480	9.6%	73.8

About one customer in ten said yes. That is our baseline, so any group above 9.6% is doing better than average.

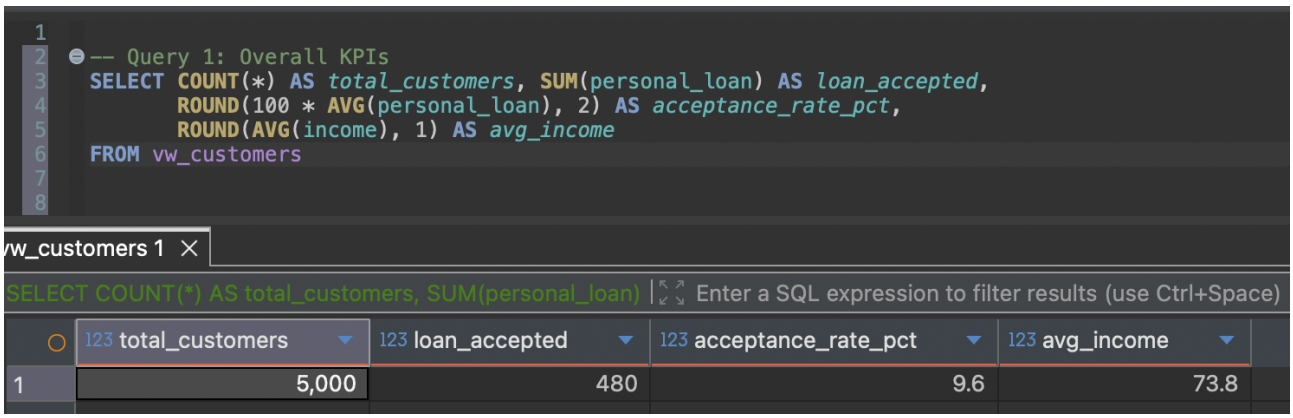


Figure 7: Query 1 and its result.

Query 2: Acceptance by income band

Income band	Customers	Accepted	Acceptance rate
Low (under 50K)	1,869	0	0%
Medium (50-99K)	1,909	41	2.15%
High (100-149K)	769	219	28.48%
Very High (150K+)	453	220	48.57%

Income is the clearest signal in the data. Nobody under 50K accepted, and the rate keeps climbing as income goes up.

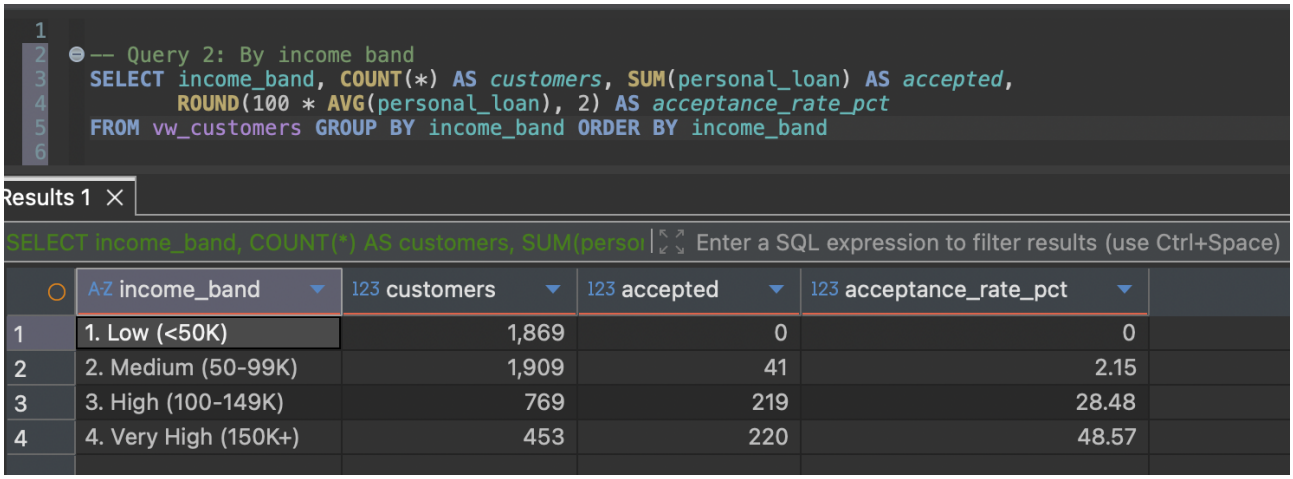


Figure 8: Query 2 and its result.

Query 3: Acceptance by education

Education	Customers	Accepted	Acceptance rate
Advanced/Professional	1,501	205	13.66%
Graduate	1,403	182	12.97%
Undergraduate	2,096	93	4.44%

Graduate and Advanced/Professional customers accept about three times as often as undergraduates, and those two groups are very close to each other.

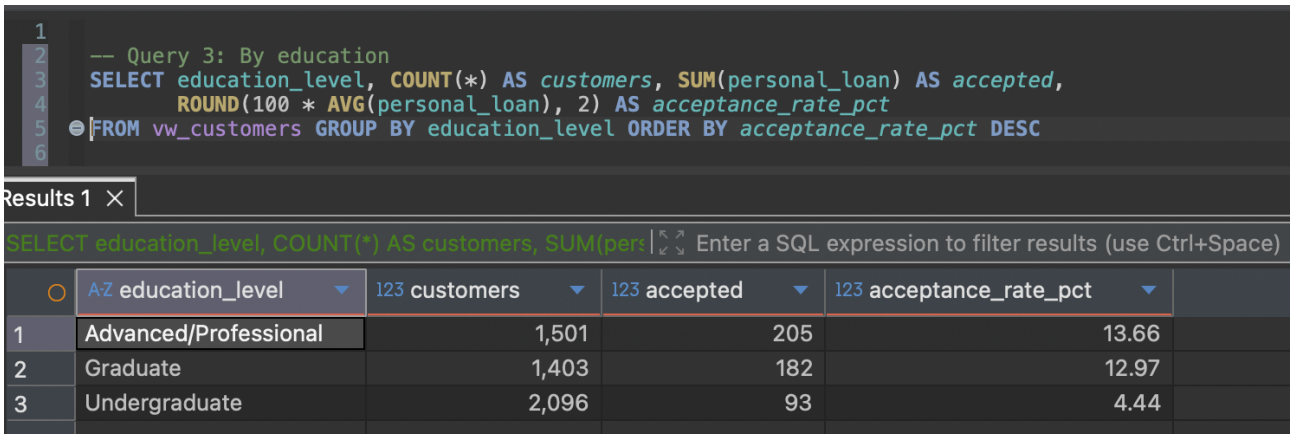


Figure 9: Query 3 and its result.

Query 4: Acceptance by CD account

CD account	Customers	Accepted	Acceptance rate
No	4,698	340	7.24%
Yes	302	140	46.36%

Only 302 customers hold a CD account, but 140 of them took the loan. That is 46.36% against 7.24% for everyone else.

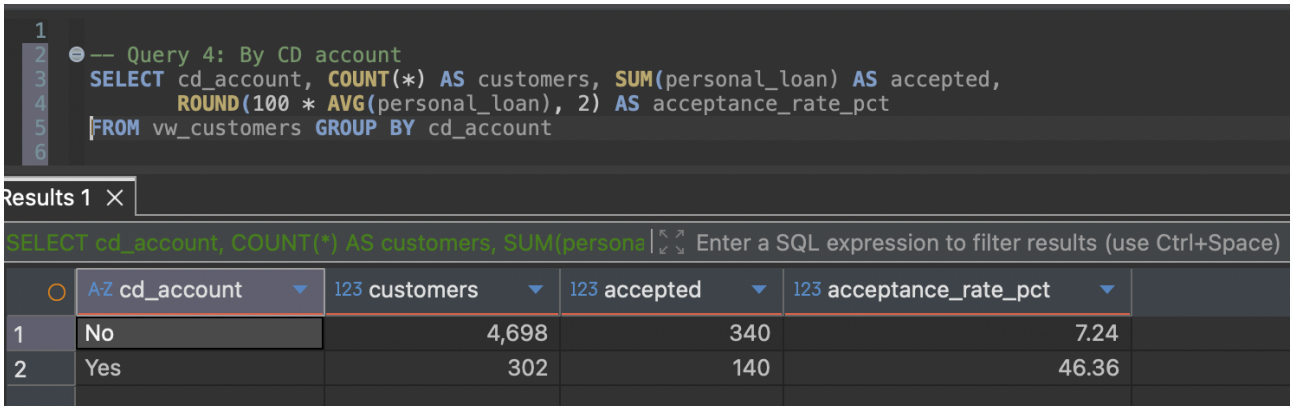


Figure 10: Query 4 and its result.

Query 5: Acceptance by card spending

Monthly card spend	Customers	Accepted	Acceptance rate
Low (under 1K)	1,683	47	2.79%
Medium (1-3K)	2,415	101	4.18%
High (3K+)	902	332	36.81%

Spending below 3K a month barely moves the needle. Above 3K the rate jumps to 36.81%, so there is a clear threshold.

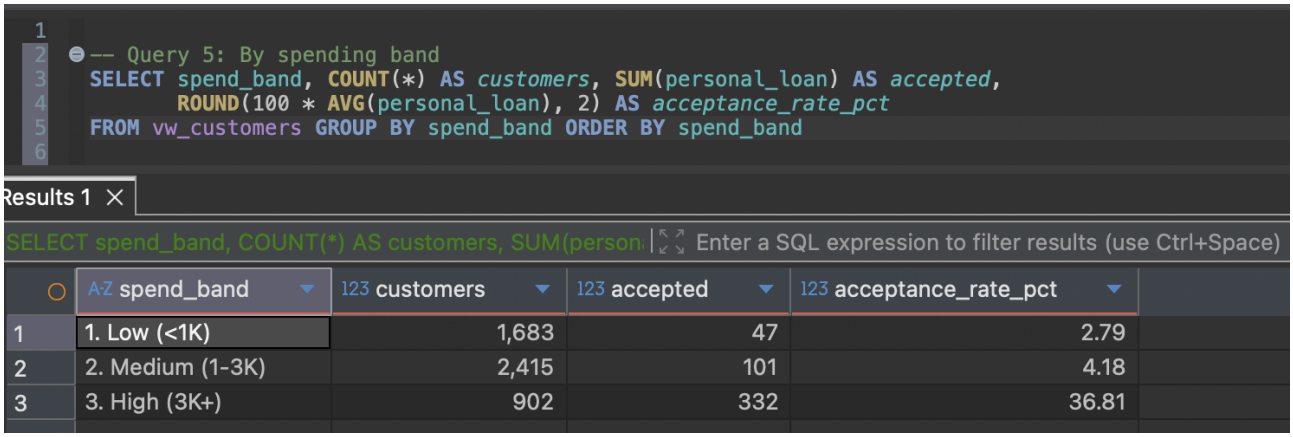


Figure 11: Query 5 and its result.

Query 6: Acceptance by family size

Family size	Customers	Accepted	Acceptance rate
1	1,472	107	7.27%
2	1,296	106	8.18%
3	1,010	133	13.17%
4	1,222	134	10.97%

Larger families accept a bit more. Size 3 is highest at 13.17% and size 4 follows, while sizes 1 and 2 stay around 7% to 8%. It is a smaller effect than income.

```sql
-- Query 6: By family size
SELECT family, COUNT(*) AS customers, SUM(personal_loan) AS accepted,
       ROUND(100 * AVG(personal_loan), 2) AS acceptance_rate_pct
FROM vw_customers GROUP BY family ORDER BY family
```

bank_original 1

SELECT family, COUNT(*) AS customers, SUM(personal_loan Enter a SQL expression to filter results (use Ctrl+Space)

	123 family	123 customers	123 accepted	123 acceptance_rate_pct
1	1	1,472	107	7.27
2	2	1,296	106	8.18
3	3	1,010	133	13.17
4	4	1,222	134	10.97

Figure 12: Query 6 and its result.

Query 7: Best segments (CTE with RANK)

This query groups customers by income band and education level. It keeps only groups with at least 30 customers, so tiny groups do not distort the picture, and then ranks them by acceptance rate. It uses a common table expression (CTE) and the RANK() window function. The result was exported as top_segments.csv for the dashboard.

Rank	Income band	Education	Customers	Accepted	Rate
1	Very High (150K+)	Advanced/Professional	102	102	100%
1	Very High (150K+)	Graduate	84	84	100%
3	High (100-149K)	Graduate	133	84	63.16%
4	High (100-149K)	Advanced/Professional	147	90	61.22%
5	Very High (150K+)	Undergraduate	267	34	12.73%
6	High (100-149K)	Undergraduate	489	45	9.2%
7	Medium (50-99K)	Graduate	545	14	2.57%
8	Medium (50-99K)	Advanced/Professional	632	13	2.06%
9	Medium (50-99K)	Undergraduate	732	14	1.91%
10	Low (under 50K)	Undergraduate	608	0	0%
10	Low (under 50K)	Graduate	641	0	0%
10	Low (under 50K)	Advanced/Professional	620	0	0%

Two segments reached 100%: customers earning 150K or more with a Graduate or Advanced degree. That is 186 customers and every one of them accepted. The next best group, 100-149K with a degree, sits around 62%. Every group under 50K is at zero.

```sql
-- Query 7: Best segments (CTE + RANK window function)
WITH seg AS (
    SELECT income_band, education_level,
           COUNT(*) AS customers,
           SUM(personal_loan) AS accepted,
           ROUND(100 * AVG(personal_loan), 2) AS acceptance_rate_pct
    FROM vw_customers
    GROUP BY income_band, education_level
    HAVING COUNT(*) >= 30
)
SELECT *, RANK() OVER (ORDER BY acceptance_rate_pct DESC) AS rank_no
FROM seg;

-- Query 8: Share of all accepted loans by income band
SELECT income_band, SUM(personal_loan) AS loans,
       ROUND(100 * SUM(personal_loan) / SUM(SUM(personal_loan)) OVER (), 2) AS pct_of_all_loans
FROM vw_customers
GROUP BY income_band
ORDER BY income_band

-- Query 9: Target profile test (the campaign recommendation)
SELECT 'High income + Grad/Prof + High spend' AS profile,
       COUNT(*) AS customers, SUM(personal_loan) AS accepted,
       ROUND(100 * AVG(personal_loan), 2) AS acceptance_rate_pct
FROM vw_customers
WHERE income >= 100 AND education_level <> 'Undergraduate' AND ccavg >= 3
UNION ALL
SELECT 'High income + High spend + CD account',
       COUNT(*), SUM(personal_loan), ROUND(100 * AVG(personal_loan), 2)
FROM vw_customers
WHERE income >= 100 AND ccavg >= 3 AND cd_account = 'Yes'
UNION ALL
SELECT 'High income only (100K+)',
       COUNT(*), SUM(personal_loan), ROUND(100 * AVG(personal_loan), 2)
FROM vw_customers
WHERE income >= 100

-- Query 10: Age group (shows whether age matters)
SELECT age_group, COUNT(*) AS customers, SUM(personal_loan) AS accepted,
```

seg 1

WITH seg AS (SELECT income_band, education_level, COUN | Enter a SQL expression to filter results (use Ctrl+Space)

	income_band	education_level	customers	accepted	acceptance_rate_pct	rank_no
1	4. Very High (150K+)	Advanced/Professional	102	102	100	1
2	4. Very High (150K+)	Graduate	84	84	100	1
3	3. High (100-149K)	Graduate	133	84	63.16	3
4	3. High (100-149K)	Advanced/Professional	147	90	61.22	4
5	4. Very High (150K+)	Undergraduate	267	34	12.73	5
6	3. High (100-149K)	Undergraduate	489	45	9.2	6
7	2. Medium (50-99K)	Graduate	545	14	2.57	7
8	2. Medium (50-99K)	Advanced/Professional	632	13	2.06	8
9	2. Medium (50-99K)	Undergraduate	732	14	1.91	9
10	1. Low (<50K)	Undergraduate	608	0	0	10
11	1. Low (<50K)	Graduate	641	0	0	10
12	1. Low (<50K)	Advanced/Professional	620	0	0	10

Figure 13: Query 7 (CTE with RANK) and its ranked result.

Query 8: Where the accepted loans come from

Income band	Loans	Share of all loans
Low (under 50K)	0	0%
Medium (50-99K)	41	8.54%
High (100-149K)	219	45.63%
Very High (150K+)	220	45.83%

Customers earning 100K or more produced 439 of the 480 loans, about 91%. The bank gets almost all of its results from this group.

```sql
-- Query 7: Best segments (CTE + RANK window function)
WITH seg AS (
    SELECT income_band, education_level,
           COUNT(*) AS customers,
           SUM(personal_loan) AS accepted,
           ROUND(100 * AVG(personal_loan), 2) AS acceptance_rate_pct
    FROM vw_customers
    GROUP BY income_band, education_level
    HAVING COUNT(*) >= 30
)
SELECT *, RANK() OVER (ORDER BY acceptance_rate_pct DESC) AS rank_no
FROM seg;

-- Query 8: Share of all accepted loans by income band
SELECT income_band, SUM(personal_loan) AS loans,
       ROUND(100 * SUM(personal_loan) / SUM(SUM(personal_loan)) OVER (), 2) AS pct_of_all_loans
FROM vw_customers
GROUP BY income_band
ORDER BY income_band

-- Query 9: Target profile test (the campaign recommendation)
SELECT 'High income + Grad/Prof + High spend' AS profile,
       COUNT(*) AS customers, SUM(personal_loan) AS accepted,
       ROUND(100 * AVG(personal_loan), 2) AS acceptance_rate_pct
FROM vw_customers
WHERE income >= 100 AND education_level <> 'Undergraduate' AND ccavg >= 3
UNION ALL
SELECT 'High income + High spend + CD account',
       COUNT(*), SUM(personal_loan), ROUND(100 * AVG(personal_loan), 2)
FROM vw_customers
WHERE income >= 100 AND ccavg >= 3 AND cd_account = 'Yes'
UNION ALL
SELECT 'High income only (100K+)',
       COUNT(*), SUM(personal_loan), ROUND(100 * AVG(personal_loan), 2)
FROM vw_customers
WHERE income >= 100

-- Query 10: Age group (shows whether age matters)
SELECT age_group, COUNT(*) AS customers, SUM(personal_loan) AS accepted,
```

Results 1

Enter a SQL expression to filter results (use Ctrl+Space)

	income_band	loans	pct_of_all_loans
1	1. Low (<50K)	0	0
2	2. Medium (50-99K)	41	8.54
3	3. High (100-149K)	219	45.63
4	4. Very High (150K+)	220	45.83

Figure 14: Query 8 and its result.

Query 9: Testing the target profile

Here we tested the profile we want to recommend, next to two simpler alternatives.

Profile	Customers	Accepted	Acceptance rate
High income + Graduate/Professional + High spend	253	242	95.65%
High income + High spend + CD account	109	90	82.57%
High income only (100K+)	1,222	439	35.92%

Each extra condition makes the group smaller but much more likely to accept. Income alone gives 35.92%, and adding a graduate degree and high card spend takes it to 95.65%. The CD account profile is strong too (82.57%), but it covers only 109 customers, so we recommend the first profile.

```sql
-- Query 7: Best segments (CTE + RANK window function)
WITH seg AS (
    SELECT income_band, education_level,
           COUNT(*) AS customers,
           SUM(personal_loan) AS accepted,
           ROUND(100 * AVG(personal_loan), 2) AS acceptance_rate_pct
    FROM vw_customers
    GROUP BY income_band, education_level
    HAVING COUNT(*) >= 30
)
SELECT *, RANK() OVER (ORDER BY acceptance_rate_pct DESC) AS rank_no
FROM seg;

-- Query 8: Share of all accepted loans by income band
SELECT income_band, SUM(personal_loan) AS loans,
       ROUND(100 * SUM(personal_loan) / SUM(SUM(personal_loan)) OVER (), 2) AS pct_of_all_loans
FROM vw_customers
GROUP BY income_band
ORDER BY income_band

-- Query 9: Target profile test (the campaign recommendation)
SELECT 'High income + Grad/Prof + High spend' AS profile,
       COUNT(*) AS customers, SUM(personal_loan) AS accepted,
       ROUND(100 * AVG(personal_loan), 2) AS acceptance_rate_pct
FROM vw_customers
WHERE income >= 100 AND education_level <> 'Undergraduate' AND ccavg >= 3
UNION ALL
SELECT 'High income + High spend + CD account',
       COUNT(*), SUM(personal_loan), ROUND(100 * AVG(personal_loan), 2)
FROM vw_customers
WHERE income >= 100 AND ccavg >= 3 AND cd_account = 'Yes'
UNION ALL
SELECT 'High income only (100K+)',
       COUNT(*), SUM(personal_loan), ROUND(100 * AVG(personal_loan), 2)
FROM vw_customers
WHERE income >= 100

-- Query 10: Age group (shows whether age matters)
SELECT age_group, COUNT(*) AS customers, SUM(personal_loan) AS accepted,
```

Results 1

Enter a SQL expression to filter results (use Ctrl+Space)

	profile	customers	accepted	acceptance_rate_pct
1	High income + Grad/Prof + High spend	253	242	95.65
2	High income + High spend + CD account	109	90	82.57
3	High income only (100K+)	1,222	439	35.92

Figure 15: Query 9 and its result.

Query 10: Does age matter?

Age group	Customers	Accepted	Acceptance rate
23-29	488	49	10.04%
30-39	1,247	127	10.18%
40-49	1,257	117	9.31%
50-59	1,334	118	8.85%
60+	674	69	10.24%

Acceptance stays between 8.85% and 10.24% in every age group, right around the 9.6% average. Age does not help the bank pick who to contact.

```sql
            COUNT(*) AS customers,
            SUM(personal_loan) AS accepted,
            ROUND(100 * AVG(personal_loan), 2) AS acceptance_rate_pct
    FROM vw_customers
    GROUP BY income_band, education_level
    HAVING COUNT(*) >= 30
)
SELECT *, RANK() OVER (ORDER BY acceptance_rate_pct DESC) AS rank_no
FROM seg;

-- Query 8: Share of all accepted loans by income band
SELECT income_band, SUM(personal_loan) AS loans,
       ROUND(100 * SUM(personal_loan) / SUM(SUM(personal_loan)) OVER (), 2) AS pct_of_all_loans
FROM vw_customers
GROUP BY income_band
ORDER BY income_band

-- Query 9: Target profile test (the campaign recommendation)
SELECT 'High income + Grad/Prof + High spend' AS profile,
       COUNT(*) AS customers, SUM(personal_loan) AS accepted,
       ROUND(100 * AVG(personal_loan), 2) AS acceptance_rate_pct
FROM vw_customers
WHERE income >= 100 AND education_level <> 'Undergraduate' AND ccavg >= 3
UNION ALL
SELECT 'High income + High spend + CD account',
       COUNT(*), SUM(personal_loan), ROUND(100 * AVG(personal_loan), 2)
FROM vw_customers
WHERE income >= 100 AND ccavg >= 3 AND cd_account = 'Yes'
UNION ALL
SELECT 'High income only (100K+)',
       COUNT(*), SUM(personal_loan), ROUND(100 * AVG(personal_loan), 2)
FROM vw_customers
WHERE income >= 100

-- Query 10: Age group (shows whether age matters)
SELECT age_group, COUNT(*) AS customers, SUM(personal_loan) AS accepted,
       ROUND(100 * AVG(personal_loan), 2) AS acceptance_rate_pct
FROM vw_customers
GROUP BY age_group
ORDER BY age_group
```

Results 1

SELECT age_group, COUNT(*) AS customers, SUM(personal | Enter a SQL expression to filter results (use Ctrl+Space)

	age_group	customers	accepted	acceptance_rate_pct
1	23-29	488	49	10.04
2	30-39	1,247	127	10.18
3	40-49	1,257	117	9.31
4	50-59	1,334	118	8.85
5	60+	674	69	10.24

Figure 16: Query 10 and its result.

6. Power BI Dashboard

We loaded the clean view and the top segments table into Power BI as CSV files. Row counts matched the SQL results (5,000 customers).

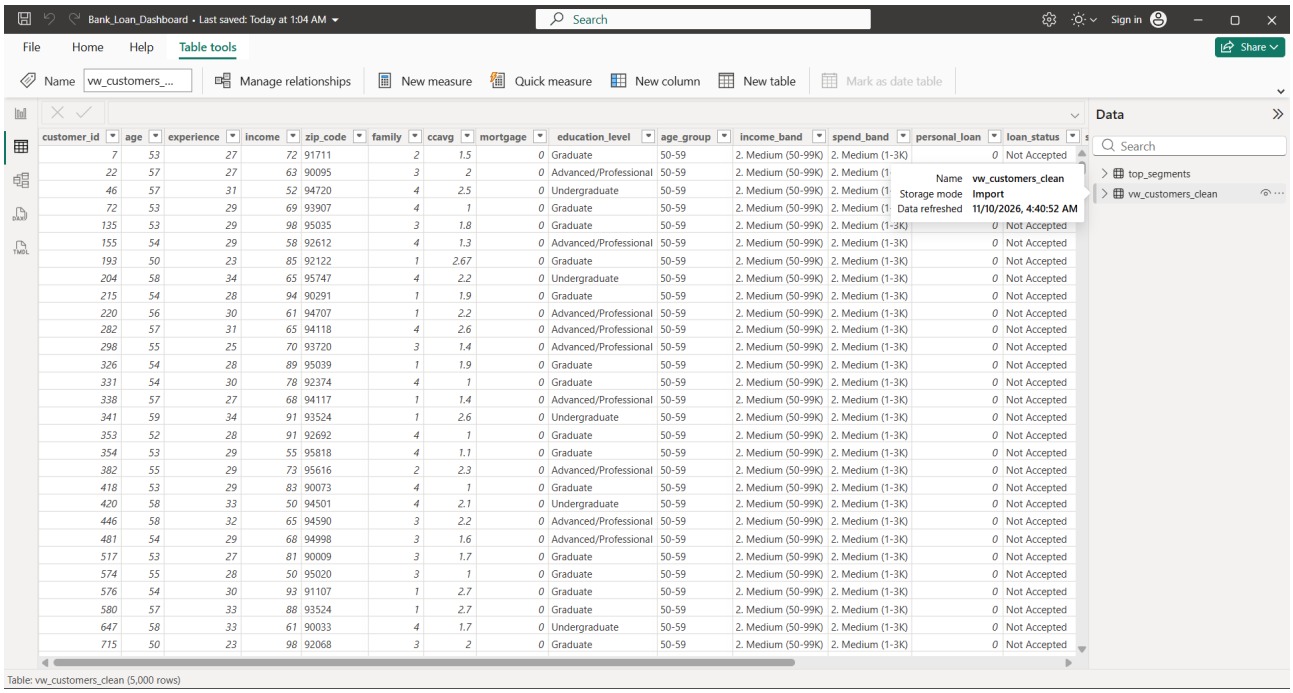


Figure 17: The clean data loaded in Power BI (table view).

We created five DAX measures:

- **Total Customers** = COUNTROWS(vw_customers_clean)
- **Loan Customers** = CALCULATE(COUNTROWS(vw_customers_clean), vw_customers_clean[personal_loan] = 1)
- **Acceptance Rate** = DIVIDE([Loan Customers], [Total Customers], 0)
- **Avg Income** = AVERAGE(vw_customers_clean[income])
- **Avg CC Spend** = AVERAGE(vw_customers_clean[ccavg])

The dashboard has three pages, and the age group and family size slicers let the viewer filter the results.

Page 1: Overview

Four cards show 5,000 customers, 480 loan customers, a 9.60% acceptance rate and an average income of 73.77. A donut chart shows the 9.6% accepted and 90.4% not accepted split, and two charts show acceptance by income band and by education level.

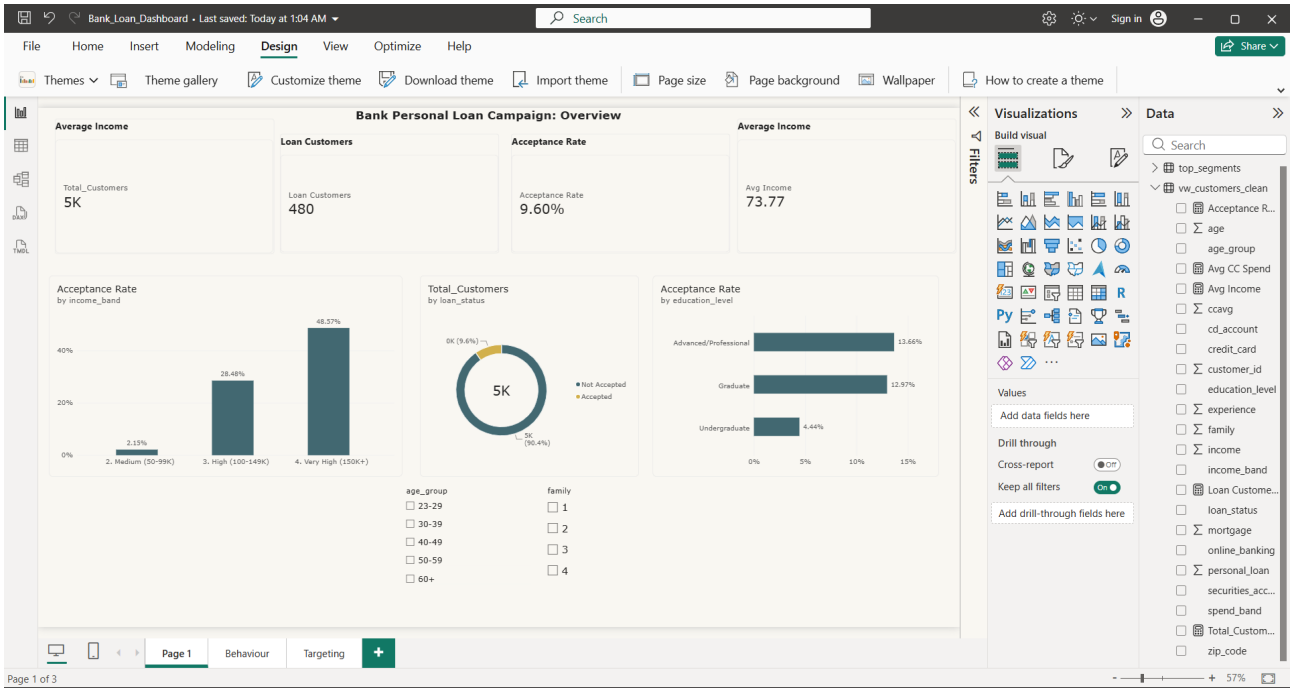


Figure 18: Dashboard page 1, Overview.

Takeaway: acceptance rises sharply with income, from 0% under 50K to 48.57% at 150K and above.

Page 2: Behaviour

Column charts compare acceptance by CD account, card spend, online banking and credit card. A scatter plot places each customer by income and card spend, and a colour-coded matrix shows acceptance by income band and education level.

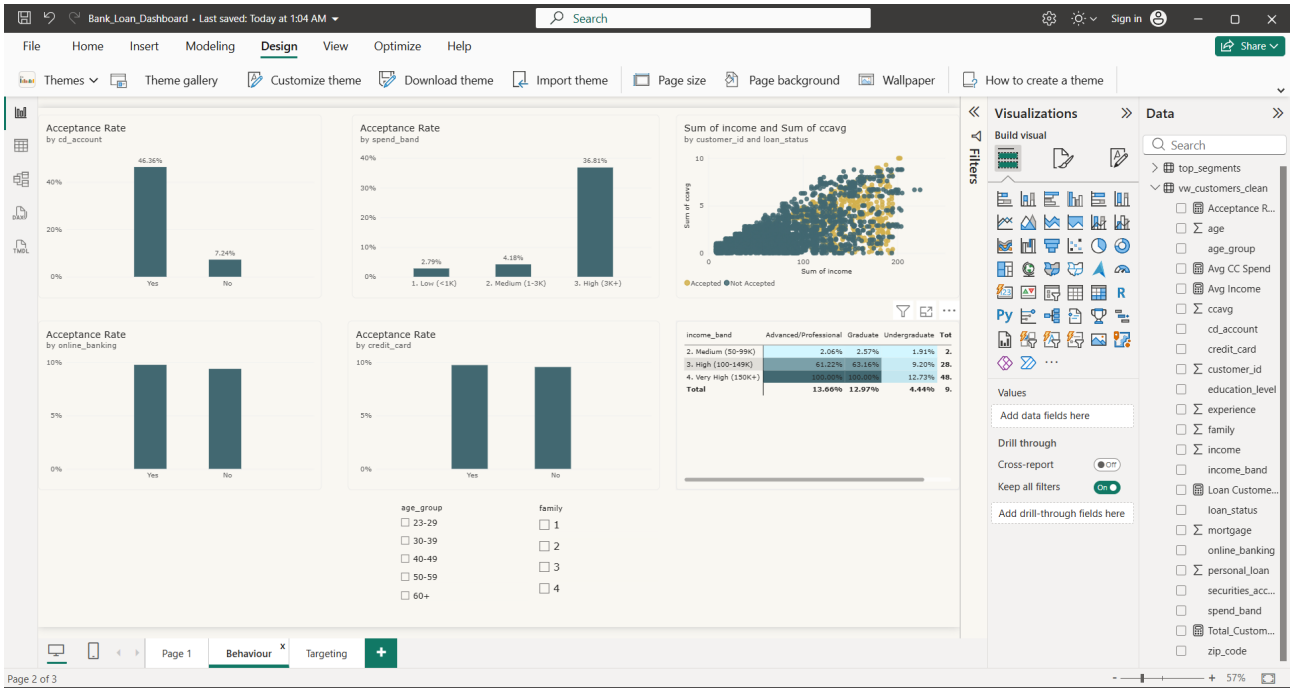


Figure 19: Dashboard page 2, Behaviour.

Takeaway: CD account holders and high card spenders accept far more often, while online banking and credit card ownership make no difference.

Page 3: Targeting

A table ranks the customer segments, and next to it sit the recommendations and the key insight about the best profile.

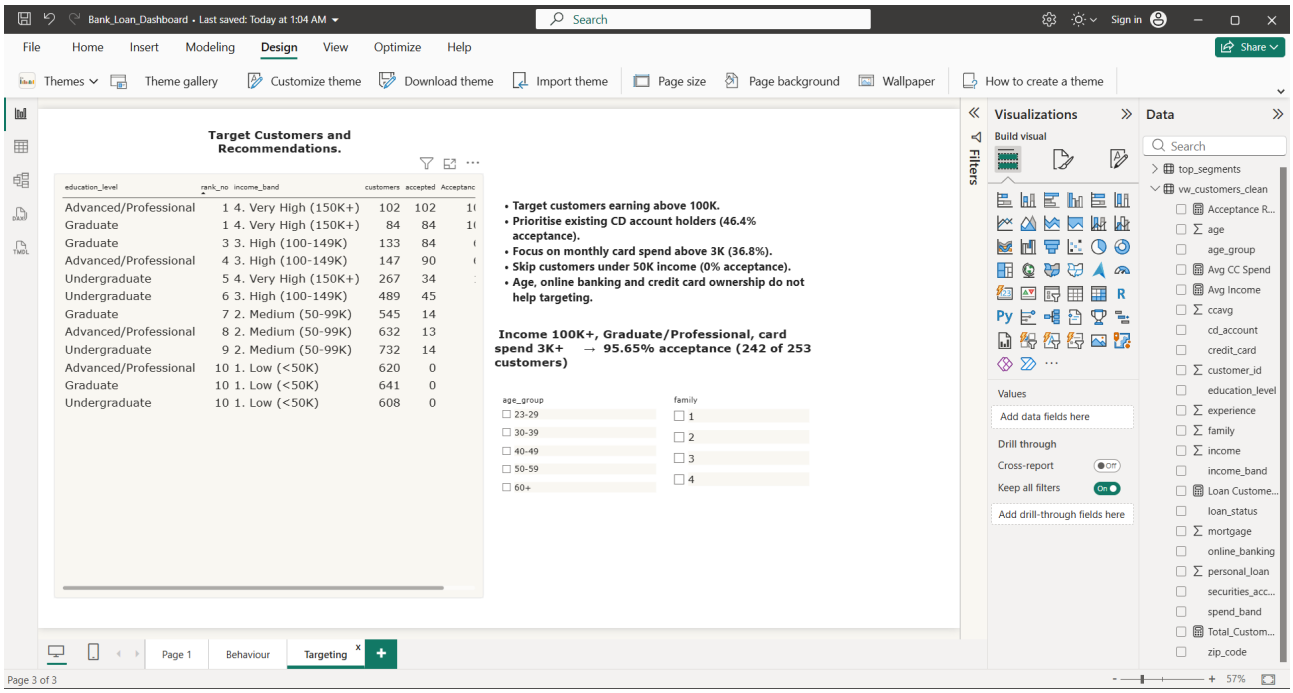


Figure 20: Dashboard page 3, Targeting.

Takeaway: customers with income of 150K or more and a Graduate or Advanced degree accepted every time (100%).

7. Key Insights

Factor	Finding
Income	The strongest driver. Nobody under 50K accepted, while 48.57% of customers at 150K+ did. Customers earning 100K or more produced about 91% of all accepted loans (439 of 480).
Education	Graduate (12.97%) and Advanced/Professional (13.66%) customers accept about three times as often as undergraduates (4.44%).
CD account	Holders accept at 46.36% against 7.24% for others, about six times higher.
Card spending	Customers spending 3K or more a month accept at 36.81%. Lower spenders stay below 5%.
Income and education together	At 150K+, Graduate and Advanced/Professional customers both reach 100%. At 100-149K they reach 63.16% and 61.22%.
Family size	A small effect. Size 3 is highest at 13.17% and size 4 is 10.97%, while sizes 1 and 2 are around 7% to 8%.
Online banking and credit card	Acceptance stays around 9% to 11% either way, so neither helps with targeting.
Age	No effect. Every age group sits between 8.85% and 10.24%, close to the 9.6% average.

8. Recommendation

The best segment we found is customers with **income of 100K or more, a Graduate or Advanced/Professional degree, and monthly card spend of 3K or more**. This group has 253

customers, and 242 of them accepted, an acceptance rate of **95.65%**.

That segment is only about 5% of the customer base (253 of 5,000), yet it holds about half of everyone who accepted (242 of 480). Contacting it first would give the bank most of the benefit at a small fraction of the cost. For context, customers earning 100K or more produced 439 of the 480 accepted loans, about 91%.

What we recommend for the next campaign:

1. Target customers earning above 100K, especially Graduate and Advanced/Professional.
2. Prioritise existing CD account holders.
3. Focus on customers who spend above 3K a month on their card.
4. Skip customers earning under 50K, since none accepted.
5. Do not use age, online banking or credit card ownership to choose targets.

9. Limitations and Future Work

- The data covers one campaign only, so results may change in another period or market.
- The analysis is descriptive. It shows which groups accepted but does not predict individual customers.
- Several factors overlap, for example high income and high card spend, so their effects are not fully separated.

A next step would be a logistic regression or decision tree model that scores each customer's chance of accepting, tested on a new campaign.

10. Conclusion

Only 9.6% of customers accepted the personal loan, but acceptance was far higher in clear groups. Income was the strongest factor, followed by CD account ownership, card spending and education. Age, online banking and credit card ownership made no real difference. By focusing on high-income, highly educated customers who spend heavily on their cards, the bank can reach a 95.65% acceptance rate and capture about half of all likely acceptors while contacting only 5% of its customers.

Appendix

- GitHub repository: github.com/masimsiddiqiui/bank-personal-loan-sql-powerbi
- SQL scripts: the `sql` folder in the repository
- Power BI file: `powerbi/Bank_Loan_Dashboard.pbix`
- Dashboard screenshots: `powerbi/screenshots`

Clean view definition (vw_customers)

```
CREATE OR REPLACE VIEW vw_customers AS
SELECT
    ID AS customer_id,
    Age AS age,
    ABS(Experience) AS experience,                    -- fixes negative values
    Income AS income,
    LPAD(`ZIP Code`, 5, '0') AS zip_code,             -- ZIP as 5-digit text
    Family AS family,
    CCAvg AS ccavg,
    Mortgage AS mortgage,
```

```
    CASE Education WHEN 1 THEN 'Undergraduate'
                   WHEN 2 THEN 'Graduate'
                   ELSE 'Advanced/Professional' END AS education_level,
    CASE WHEN Age < 30 THEN '23-29'
         WHEN Age < 40 THEN '30-39'
         WHEN Age < 50 THEN '40-49'
         WHEN Age < 60 THEN '50-59'
         ELSE '60+' END AS age_group,
    CASE WHEN Income < 50  THEN '1. Low (<50K)'
         WHEN Income < 100 THEN '2. Medium (50-99K)'
         WHEN Income < 150 THEN '3. High (100-149K)'
         ELSE '4. Very High (150K+)' END AS income_band,
    CASE WHEN CCAvg < 1 THEN '1. Low (<1K)'
         WHEN CCAvg < 3 THEN '2. Medium (1-3K)'
         ELSE '3. High (3K+)' END AS spend_band,
    `Personal Loan` AS personal_loan,
    CASE `Personal Loan` WHEN 1 THEN 'Accepted' ELSE 'Not Accepted' END AS loan_status,
    IF(`Securities Account` = 1, 'Yes', 'No') AS securities_account,
    IF(`CD Account` = 1, 'Yes', 'No')         AS cd_account,
    IF(Online = 1, 'Yes', 'No')               AS online_banking,
    IF(CreditCard = 1, 'Yes', 'No')           AS credit_card
FROM Bank_Original;
```